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AcousticMoEKAN: Sparse Mixture-of-KAN Experts for Time-Frequency Forecasting of OLTC Acoustic Signals

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ABSTRACT On-load tap-changers (OLTCs) regulate distribution voltage and undergo frequent mechanical operations that can accelerate wear. Acoustic monitoring is non-intrusive, but forecasting abnormal behavior directly from 44.1 kHz waveforms is computationally prohibitive for multi-step prediction. We formulate OLTC acoustic prognostics as time–frequency forecasting: each tap-changing event is converted to a short-time Fourier transform (STFT) magnitude spectrogram with 513 frequency-bins, and the task is to predict future spectrogram evolution. We propose AcousticMoEKAN, an acoustic-optimized mixture of Kolmogorov–Arnold networks (KAN) that expands the expert pool from the four analytic bases used in MoEKAN to ten complementary experts and replaces dense Transformer gating with sparse Mixture-of-Block Attention (MoBA) gating for efficient top-k expert routing. Experiments on a large-scale accelerated-aging dataset comprising 195,994 tap-changing events show that AcousticMoEKAN improves accuracy averaged over the DC component and four equal-width frequency bands, outperforming linear models, convolutional neural networks (CNNs), graph neural networks (GNNs), Transformers, and KAN baselines. AcousticMoEKAN reduces average mean squared error (MSE), mean absolute error (MAE), and mean absolute percentage error (MAPE) by 20.6%/9.5%/11.6% relative to MoEKAN and by 25.1%/11.0%/12.0% relative to the best non-KAN baselines (RLinear for MSE, STID for MAE/MAPE), with consistent gains across low and high frequency bands. These results indicate that expert-augmented KAN modeling and sparse MoBA gating provide an effective framework for OLTC acoustic condition forecasting in the STFT domain.

INDEX TERMS Deep learning, On-load tap-changer (OLTC), acoustic prognostics, STFT spectrogram forecasting, mixture-of-experts, Kolmogorov–Arnold networks (KAN), Mixture-of-Block Attention.

I. INTRODUCTION

With the proliferation of artificial intelligence (AI) services, electricity demand from data centers and AI-server infrastructure has been rising rapidly. High-density GPU servers and high-power PSUs can impose industrial-scale loads, and their consumption can fluctuate sharply as AI workloads and cloud traffic change. This variability, combined with power-electronics-based characteristics, increases sensitivity to power-quality disturbances: voltage sags can trigger outages or cooling degradation, while overvoltage can accelerate aging and failures. Accordingly, enhancing low-voltage ride-through of data-center loads

and analyzing the impacts of sags and short interruptions have attracted growing attention, highlighting the need for equipment that maintains grid voltage within a target range [1], [2].

Against this backdrop, transformer on-load tap-changers (OLTCs) are increasingly critical for distribution-voltage regulation because they switch taps under load to keep system voltage within a prescribed range. However, OLTCs are complex electromechanical systems that repeatedly combine mechanical actuation with electrical switching, and their failures can arise from degradation, wear, and arcing. Prior reviews emphasize the importance

of both offline/online monitoring and sensor-based diagnostics for predictive maintenance [3], while recent surveys indicate that non-intrusive acoustic/vibro-acoustic sensing is becoming more common in power equipment assessment [4]. Because OLTC failures can escalate to voltage instability, overload, and service outages, proactive failure prediction is essential for reliable operation [3], [4].

Selecting diagnostic data sources that reflect failure mechanisms is crucial for OLTC prognostics. Dissolved gas analysis (DGA) is effective for electrical/thermal abnormalities but can be delayed and is less sensitive to purely mechanical faults [5]. Motor current signatures during tap-changing can reveal friction, sticking, wear, and control anomalies, yet they are sensitive to temperature, supply conditions, installation, and noise, often requiring equipment-specific baselines [6]. Voltage signatures can indicate degraded voltage-regulation performance, but they also embed external grid/load effects, complicating fault attribution [7]. Temperature monitoring can track overheating but is limited by thermal inertia and sensor placement [8]. Vibration monitoring is effective for mechanical wear and impacts but is sensitive to ambient vibration and may respond weakly to arcing/contact anomalies [9].

By contrast, acoustic-signal-based diagnosis can jointly capture mechanical and electrical anomalies because contact impacts, friction, and arcing generate broadband emissions during tap-changing. Prior work has demonstrated OLTC assessment using acoustic features and distance metrics, acoustic-emission-based fault detection with machine learning, and empirical validation of acoustic analysis for OLTC abnormalities [10]–[12]. Deep-learning-based anomaly detection and OLTC acoustic monitoring frameworks further suggest that learned representations can effectively exploit acoustic data [13], [14]. Overall, acoustic sensing can capture subtle early-stage changes with low latency and is practical for field deployment due to non-contact installation [10]–[14].

Acoustic-based predictive maintenance has been extensively studied not only for OLTCs but also for a broad range of industrial assets. A review covering sound-based predictive maintenance from feature engineering to deep learning reports that acoustic signals can provide informative condition cues even in low-quality, high-noise environments [15]. Lightweight deep-learning diagnostic models under low-quality and noisy data [16], bearing fault diagnosis based on acoustic data fusion and mode decomposition [17], bearing diagnosis combining time-frequency analysis with transfer-learning CNNs [18], unsupervised acoustic-based fault diagnosis that accounts for environmental noise [19], and surveys on early fault diagnosis for induction motors using acoustics/acoustic emission [20] collectively support the effectiveness of acoustic data across rotating machinery and drive-train systems. In addition, a study analyzing the effect of

acoustic sensor type on OLTC fault classification performance shows that experimental design (e.g., sensor characteristics and mounting conditions) can directly influence diagnostic accuracy [21]. Furthermore, studies using scalograms and mel spectrograms for defect diagnosis in drilling processes [22], comparing acoustic anomaly-detection models for pumped-storage facilities [23], proposing acoustic-based diagnostic models for low-speed bearings [24], integrating audio-sensor-based diagnosis with explainable AI (LIME/SHAP) [25], optimizing features for acoustic-based fault diagnosis in milling machines [26], modeling acoustics in HVDC converter-station equipment [27], developing hierarchical CNN-based condition recognition for pumped-storage facilities [28], applying transfer learning to machine-tool fault diagnosis [29], and audio-based motor fault diagnosis using SOM-LSTM [30] suggest that converting acoustic signals into time-frequency representations and feeding them to learning models is repeatedly effective across heterogeneous equipment types. Beyond these, improvements to onset time picking in SNR signals [31], comparisons of wavelet-based features and classifiers for engine fault diagnosis [32], localization and identification in metal plates [33], and feature extraction for welding acoustic signals using synchrosqueezed transform (SST) and PCA [34] demonstrate that time-frequency features can play a critical role even under low signal-to-noise ratios or incipient faults. Overall, acoustic monitoring constitutes a versatile data source that can cover both electrical and mechanical faults, and it is often considered advantageous for capturing early abnormal signatures when combined with time-frequency features and learning-based models [15]–[34].

OLTCs operate repetitively and are therefore prone to wear-related degradation, and field statistics commonly report OLTC-related faults as a substantial fraction of transformer failures. Moreover, arcing within an OLTC can escalate into severe incidents, implying that loss of voltage-regulation function can translate directly into power-quality and operational risks. Consequently, prognostic approaches that anticipate failures before hazardous events are required, and transformer PHM studies increasingly emphasize machine-learning-based predictive maintenance directions [35].

To translate forecasts into actionable maintenance decisions, prediction horizon and lead time are key. OLTCs may perform multiple tap-changes per day, and detecting incipient signatures accumulated over repeated operations can inform maintenance planning. In practice, day-ahead scheduling motivates lead times on the order of 24 h as a useful compromise between forecast reliability and response time, consistent with prior work on predictive indicators and maintenance scheduling frameworks [36]–[38]. Hence, forecasting models that predict near-future

state evolution and connect it to operational planning are increasingly important [36]–[38].

Time-series forecasting, particularly long-horizon forecasting, goes beyond fitting historical data: it requires stable extrapolation of accumulated nonlinear changes over time, such as trend shifts, threshold phenomena, and deformations of seasonality, as well as interactions among variables. In this sense, model expressivity and inductive bias strongly influence performance. Studies on vibration-based condition monitoring and bearing prognostics for industrial equipment also repeatedly emphasize the importance of model architectures that can learn long-term trends and failure patterns from raw signals [39].

Linear models are computationally efficient and serve as strong baselines by predicting future values as linear combinations of past observations. NLinear shows that simple linear mappings can be competitive on long-horizon benchmarks [40], and related analyses indicate that normalization and input–output mapping design can drive much of the observed gains [41]. However, industrial signals often exhibit non-stationary nonlinear behavior, so purely linear models can miss localized composite fault signatures [40], [41].

Graph neural network (GNN) approaches model variables as nodes and dependencies as edges, learning interactions via message passing. STID demonstrates that simplified spatio-temporal identity mechanisms can be strong baselines [42], while TimeGNN targets time-varying relationships via dynamic-graph learning [43]. Nonetheless, these methods require defining or learning relational structures, which can be challenging when dependencies evolve or are noisy [42], [43].

CNN-based models extract local patterns efficiently and often perform well on event-driven or short-term variations. SCINet uses hierarchical splitting and interaction learning to capture multi-scale dependencies [44], and TimesNet reconstructs time series into 2D representations and applies 2D CNNs to learn intra-/inter-period variations [45]. However, capturing very long-range context with convolutions can require deeper networks or more elaborate designs [44], [45].

Transformer-family models use self-attention to model long-range dependencies, which is advantageous for long-context forecasting. PatchTST reduces attention cost by tokenizing inputs into patches [46], while iTransformer redefines tokens from time steps to variables to better capture inter-variable dependencies [47]. Nevertheless, attention cost grows rapidly with sequence length, so practical deployment requires careful consideration of efficiency and data constraints [46], [47].

In this context, Kolmogorov-Arnold Networks (KANs) provide a new paradigm in which each connection implements a learnable one-dimensional function, offering a flexible mechanism for approximating complex multivariate mappings [48]. Recent work has examined

KAN-based designs for time-series forecasting [49], and MoEKAN further combines KAN experts with Transformer-based gating to report competitive performance on industrial time series [50]. Overall, KAN-based mixture-of-experts structures can decompose heterogeneous dynamics (e.g., trends, periodicity, and localized shocks) into specialized components, which is structurally well aligned with forecasting OLTC acoustic signals where nonlinearity and non-stationarity coexist [48]–[50].

Building on this background, this paper designs a learning-based model that captures early state changes from acoustic signals measured during OLTC tap-changing and forecasts the future evolution of acoustic characteristics. Specifically, we reconfigure a KAN-based MoE forecasting framework for OLTC acoustic time series and propose AcousticMoEKAN, validating its effectiveness in a power-equipment predictive-maintenance scenario through comparisons against diverse time-series forecasting baselines.

Motivated by this problem setting, we study long-horizon forecasting of OLTC tap-changing acoustic signatures for pre-failure indication. Instead of directly predicting 44.1 kHz raw waveforms, we represent each event as a compact STFT magnitude spectrogram and learn the temporal evolution of spectrum energy per frequency-bin. Prior acoustic studies have reported that OLTC degradation and faults can manifest as changes in time-frequency patterns and statistical acoustic features [10]–[14]. We extend the MoEKAN framework to design AcousticMoEKAN by (i) enlarging the set of analytically motivated KAN experts to better approximate envelope modulation and resonant structures observed in acoustic spectra, and (ii) introducing MoBA-based sparse gating that enables input-adaptive expert selection. Extensive experiments on an accelerated-aging dataset consisting of 195,994 tap-changing events demonstrate that AcousticMoEKAN consistently outperforms linear, CNN, GNN, Transformer, and existing KAN-family baselines, both in aggregate metrics and in per-frequency-band metrics.

Our main contributions are summarized as follows:

- 1) We present a scalable time-frequency forecasting formulation that converts audio-rate waveforms into fixed-length sequences of STFT spectrograms, enabling learning from long-duration recordings while preserving diagnostically meaningful spectral structure.
- 2) We propose AcousticMoEKAN, which expands the four analytically motivated experts in MoEKAN (Taylor/Wavelet/Jacobi/Fourier) into ten complementary experts specialized for oscillatory, damped, envelope-modulated, and locally transient patterns, thereby better modeling the composite shapes of OLTC acoustic spectra.

- 3) We introduce MoBA-based sparse gating for expert routing and conduct comprehensive experiments, including full-spectrum and band-wise MSE/MAE/MAPE analyses as well as qualitative comparisons between initial and pre-failure data, to verify the robustness and effectiveness of the proposed approach.

The remainder of this paper is organized as follows. Section II reviews related work on OLTC condition monitoring and time-series forecasting. Section III describes the experimental setup, data acquisition, and construction of the STFT-based dataset. Section IV presents the proposed AcousticMoEKAN architecture, including the extended expert design and MoBA-based sparse gating. Section V reports experimental results and additional analyses. Finally, Section VI concludes the paper and discusses future research directions.

II. RELATED WORK

In this section, we summarize the data sources (sensor signals) that have been used in prior studies for OLTC failure prediction and diagnosis, and we review representative deep learning and machine learning network architectures for analyzing and forecasting such signals.

A. Data Sources and Time–Frequency Representations for OLTC Condition Monitoring

Dissolved gas analysis (DGA) is a traditional technique that indirectly reflects electrical/thermal abnormalities in OLTCs (e.g., arcing and overheating) through changes in dissolved gas composition (e.g., H₂, CH₄, and C₂H₂) [3]. Its limitations include delayed responsiveness until gases accumulate and low sensitivity to purely mechanical defects [3]. To improve early warning, recent work has combined DGA with time-series forecasting and anomaly detection [5].

Operation current signatures (motor/switching current waveforms during tap-changing) can more directly indicate drive-train issues such as friction increase, sticking, wear, and operation delays [6]. Although they are available for each tap-changing event, they are strongly affected by temperature, lubrication, supply variation, installation, and noise, often requiring equipment-specific baselines and calibration [6], [7].

Secondary-side operation voltage signatures provide outcome-oriented indicators of degraded voltage-control functionality (e.g., switching failure/delay or inability to maintain target voltage) [7]. However, voltage deviations are also influenced by external load and grid conditions, making it difficult to localize internal causes using voltage alone [7].

Temperature monitoring tracks thermal variations (e.g., tank/bushing) to follow overheating driven by increased contact resistance or frictional heating, and its

applicability/limitations have been reviewed in the literature [8]. In practice, thermal inertia can reduce sensitivity to early-stage anomalies, and localized defects may be missed depending on sensor placement [8].

Vibration analysis using accelerometers is effective for diagnosing mechanical faults (wear, impacts, imbalance) in OLTC components and has been widely used in practice [4]. Prior work also reports small-sample OLTC fault identification using time–frequency vibration features with lightweight classifiers [9]. Nevertheless, vibration measurements are sensitive to ambient vibration and mounting conditions and may respond weakly to electrical arcing/contact anomalies [3], [4].

Acoustic signals (audible/ultrasonic/acoustic emission) provide a non-intrusive modality that can jointly reflect mechanical and electrical phenomena, because contact impacts, friction, and arcing generate broadband emissions during tap-changing [4], [10]. Existing studies include feature/distance-based monitoring, machine-learning-assisted fault detection, feasibility assessments, and deep anomaly-detection frameworks (e.g., autoencoders/VAEs) [10]–[14]. However, acoustic diagnosis can be sensitive to ambient noise and sensor configuration, so signal quality control and data standardization remain practical challenges [21].

Because OLTC acoustics are inherently non-stationary, multiple short-duration events (impulses, friction components, and arcing-induced high-frequency content) can overlap during tap-changing. Purely time-domain features may therefore be insufficient, and time–frequency representations such as spectrograms are commonly adopted to separate band-wise dynamics [4], [9], [10].

Time–frequency transforms (e.g., STFT, wavelets, and SST) support band-specific energy tracking and temporal localization of fault events, and many studies report improved fault classification/anomaly detection when such representations are used [11], [12], [14], [18], [22]. More broadly, surveys on acoustic predictive maintenance emphasize time–frequency feature extraction as a key component, including lightweight deep-learning designs for robustness under noisy and low-quality conditions [15], [16].

B. Network Architectures for Predictive Maintenance and Time-Series Forecasting

Predictive maintenance requires forecasting future risk-state transitions with a prescribed lead time rather than only classifying the current condition. Long-horizon forecasting is challenging due to nonlinear evolution, distribution shift, and error accumulation, and performance depends on the interaction between inductive bias and representational capacity [40], [46]. Accordingly, we review representative forecasting architectures spanning Linear, CNN, GNN, Transformer, and KAN families.

Linear models forecast future values as linear combinations of past observations and remain strong, low-cost baselines. NLinear shows competitive long-horizon performance with channel-wise linear mappings combined with normalization/denormalization [40], and RLinear argues that normalization and linear transforms can explain many gains previously attributed to complex modules [41]. However, strict linearity limits modeling of nonlinear interactions and threshold-like behaviors in real-world signals [40].

GNN-based models represent variables as graph nodes and learn dependencies via message passing. STID proposes a simplified baseline using node/time ID embeddings, showing that heavy spatio-temporal graph structures are not always necessary for strong performance [42]. TimeGNN further targets time-varying relationships by learning dynamic graphs and reports improved adaptability and forecasting accuracy under evolving dependencies [43].

CNN-based models capture local patterns effectively through convolution and can extend their receptive field using dilation or hierarchical designs. SCINet learns multi-scale interactions via recursive splitting and recombination to capture longer dependencies [44]. TimesNet estimates dominant periods (FFT-based) and reconstructs time series into 2D variation tensors to learn intra-/inter-period dynamics, demonstrating generality across tasks [45].

Transformer-family models leverage self-attention for long-context dependency modeling. PatchTST reduces attention burden by patching the input and learns patch-level representations using channel-wise independent Transformer encoders [46]. iTransformer inverts tokenization (variables as tokens) to efficiently model inter-variable correlations under long inputs [47]. Despite their effectiveness, attention-based models can incur significant compute/memory costs with increasing sequence length, which must be considered for practical deployment [40], [46].

KANs (Kolmogorov-Arnold Networks) implement learnable one-dimensional functions on edges, providing adaptive nonlinear approximation motivated by the

Kolmogorov-Arnold representation theorem [48]. KAN4TSF systematically studies KAN applicability and design factors for time-series forecasting [49]. Building KANs as mixture-of-experts (MoE) further enables specialization across heterogeneous patterns (trend, periodicity, and localized events); MoEKAN combines multi-scale decomposition with Transformer-based gating to select and mix KAN experts and reports competitive performance on industrial time series [50].

III. MATERIAL

A. Experimental Setup and Accelerated Wear-test Conditions

The experiments were conducted using an ECOTAP VPD® test platform to realistically emulate the operating environment of an on-load tap-changer (OLTC) under laboratory conditions. A three-phase configuration was adopted to reproduce transformer operating conditions, and a load current of 12 A was applied to each phase. The supply voltage was set to 90~120 V, while the target OLTC voltage was maintained at 110 V. The ECOTAP controller was configured to trigger an automatic tap-change whenever the deviation from the target voltage exceeded $\pm 5\%$.

To comply with the manufacturer (MR) warranty conditions, air rather than insulating oil was used as the dielectric medium. This choice enables frequent tap operations in a non-oil environment to accelerate wear of contacts and mechanical components and to emulate an accelerated-aging scenario that can arise in practical operation. A failure event was observed after a total of 195,994 tap-changes under the air condition, and the test was conducted from September 7, 2023, to April 16, 2025. The experimental setup is shown in Fig. 1. When interpreting and extending the results of the air-based accelerated wear-test to an oil-immersed environment, one should consider that acoustic propagation characteristics may differ owing to differences in acoustic impedance, wave speed, and attenuation between oil and the steel tank.

B. Acoustic Data Acquisition and Implementation Details

Acoustic signals generated during OLTC operations were captured using a Brüel & Kjær Type 4955 microphone. The sensor provides a 10 Hz~20 kHz frequency response and a low noise floor of approximately 6.5 dBA, enabling accurate capture of subtle acoustic variations during tap transitions. Signals were recorded at a 44.1 kHz sampling rate to preserve spectral content, yielding 44,100 samples per 1-s segment. The proposed model was implemented in PyTorch 2.0 with CUDA 11.7 and trained on a workstation equipped with a single NVIDIA RTX 3090 GPU (24 GB memory) and an Intel Core i9-class CPU.

Although the recordings are sampled at 44.1 kHz (44,100 samples/s), subsequent analysis is performed on fixed-



FIGURE 1. OLTC Experimental Environment.

length event-centered clips. Specifically, each OLTC tap-changing operation is represented by a 2-s segment (1 s pre-event and 1 s post-event), resulting in 88,200 samples per event, and we convert each clip into a time-frequency representation using the short-time Fourier transform (STFT). Unless otherwise stated, the STFT is computed with an FFT size of 1024 using a Hann analysis window of 1024 samples (approximately 23.2 ms at 44.1 kHz) and a hop size of 512 samples (approximately 11.6 ms), corresponding to 50% overlap between adjacent frames.

To ensure a consistent input shape across all events, STFT framing is implemented with zero padding of 512 samples at both boundaries (centered framing) and additional end padding of 376 samples so that the total padded length becomes 89,600 samples and aligns to an integer number of hops. With this configuration, each event is converted into a fixed-size, one-sided log-power spectrogram with 513 frequency bins spanning 0~22.05 kHz (43.07 Hz bin spacing) and 174 time frames. Spectral power is logarithmically compressed and represented on a dB scale using a stabilization constant $\epsilon = 1e-12$ to avoid numerical issues near the noise floor and to improve optimization stability under the wide dynamic range of OLTC acoustics.

Window selection is critical because the downstream forecasting task treats each frequency bin as an individual trajectory; thus, spectral leakage can artificially mix narrowband energy into adjacent bins and corrupt frequency-specific dynamics. In our recordings, a tone-like component around 12.8 kHz is frequently observed in the spectrogram, and an inappropriate analysis window can smear this component over neighboring frequencies. To quantify leakage in a model-independent manner, we compared Rectangular, Hamming, and Hann windows using a 2-s synthetic sinusoid at 12.8 kHz, chosen intentionally as a non-bin-centered tone under the 1024-point FFT at 44.1 kHz (i.e., a standard stress case for leakage). Leakage was quantified as the maximum sidelobe level outside a ± 500 Hz exclusion band around the tone, reported in dB relative to the main spectral peak, as summarized in Fig. 2. Under this criterion, the Hann

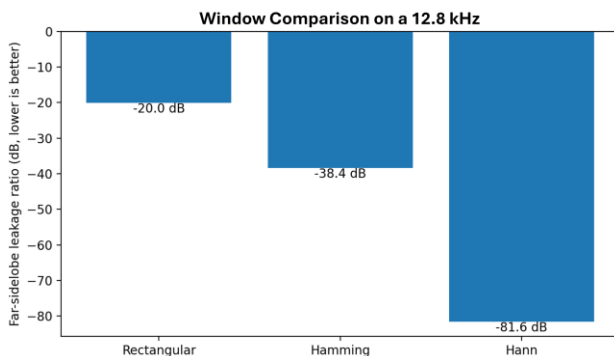


FIGURE 2. Quantitative Window Comparison.

window yielded the lowest far-frequency leakage and was therefore adopted for all preprocessing.

The overlap ratio determines the temporal sampling density of the spectrogram and directly affects both sequence length and the smoothness of frame-wise energy trajectories. We evaluated 25%, 50%, and 75% overlap (hop sizes of 768, 512, and 256 samples, respectively), which produce 116, 174, and 346 frames per 2-s event under the same padding policy. For a representative tap-changing event, 25% and 50% overlap produced visually similar frame-wise energy trajectories, whereas 75% overlap led to noticeably more jagged frame-to-frame fluctuations. To express this difference numerically, we computed a simple roughness score defined as the mean absolute frame-to-frame change of the normalized full-band energy curve (higher indicates greater temporal fluctuation). The roughness scores were 14.88 (25%), 15.96 (50%), and 21.40 (75%). Compared to 50% overlap, 25% overlap differed by only 6.8%, whereas 75% overlap increased roughness by 34.1% while nearly doubling the per-event sequence length (346 vs. 174 frames). Considering both stability and computational cost, we use 50% overlap as a practical balance, as summarized in Fig. 3.

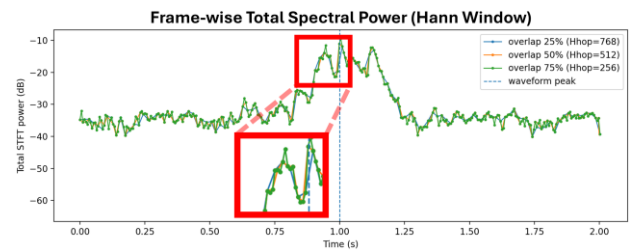


FIGURE 3. Frame-wise Spectral Energy Trajectories under Different Overlap Ratios (25%, 50%, 75%) for the Representative Event.

C. Acoustic-Signature Comparison Between Air and Oil Media

Although the accelerated wear-test was performed in an air medium to satisfy the manufacturer warranty constraints, practical transformer OLTCs operate in insulating oil; therefore, potential differences in acoustic characteristics between the two environments should be examined. To this end, representative tap-change events were selected under air and oil conditions, and we compared (i) STFT-based spectrogram patterns, (ii) energy-envelope shapes, (iii) power spectral density (PSD) distributions, and (iv) temporal energy variations across frequency bands.

As shown in the spectrogram comparison in Fig. 4, both air and oil conditions exhibit a broadband energy increase at the tap-change instant, with most energy concentrated in the low-frequency region (particularly 0~2 kHz). Under the oil condition, certain high-frequency components may be relatively attenuated or specific resonance components may appear more pronounced due to the medium's acoustic impedance and attenuation characteristics; nevertheless,

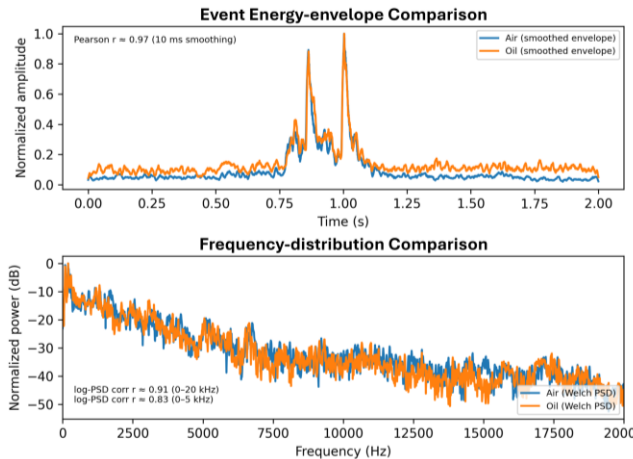


FIGURE 4. Similarity Comparison of Energy Envelopes and PSD Distributions.

the event timing and the dominant frequency region are consistent across the two media.

Because air and oil differ in acoustic impedance and attenuation, the raw waveforms are unlikely to match perfectly down to fine-scale phase and reverberation components. Accordingly, rather than using alignment-sensitive measures such as pixel-wise comparisons, we quantitatively assessed cross-medium similarity based on macroscopic characteristics of the tap-change event (i.e., energy envelopes, frequency distributions, and the simultaneity of band-limited energy bursts), as summarized in Fig. 5.

In addition, a representative-event analysis (one event per condition) confirmed strong similarity between the two media. For a 2-s event window sampled at 44.1 kHz, the Pearson correlation coefficient between the aligned analytic-amplitude envelopes was 0.97, and the correlation of the log-PSD spectra over 0~20 kHz was 0.91. The spectral centroid was 1.26 kHz for air and 1.21 kHz for oil, and the energy fraction in 0~2 kHz was 82.6% and 81.2%,

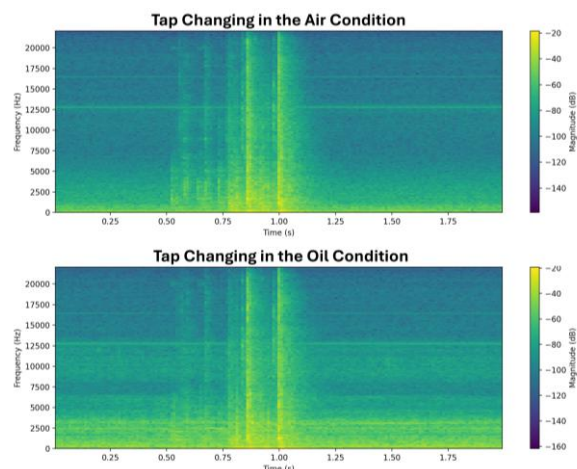


FIGURE 5. Comparison of STFT Spectrograms for Air and Oil Conditions over the 0~2-s Interval.

respectively. These results indicate that, while medium-dependent differences exist in fine resonance and high-frequency attenuation, the macroscopic temporal structure (envelope) and overall frequency distribution that characterize the tap-change event are largely preserved.

Fig. 6 compares band-wise temporal energy profiles and shows that energy bursts occur synchronously at the tap-change instant across the 100~300 Hz, 500~1000 Hz, and 2000~4000 Hz bands, with nearly identical peak timing under both conditions. However, the air condition tends to exhibit more impulsive peaks in some bands, whereas the oil condition shows smoother peaks and comparatively longer reverberation, reflecting medium-dependent attenuation and transmission characteristics.

In summary, although air and oil conditions exhibit some differences in high-frequency attenuation and resonance behavior due to medium properties, the essential temporal structure and dominant frequency distribution of tap-change events remain similar. Therefore, we expect that an acoustic model trained on the air-medium accelerated wear data will exhibit qualitatively similar behavior when applied to oil-immersed operating environments.

D. Comparative Analysis of OLTC Tap-change Event Waveforms

In this subsection, to qualitatively and quantitatively assess differences between initial and pre-failure operating conditions, we analyze representative tap-change events in terms of time-domain waveforms, amplitude distributions,

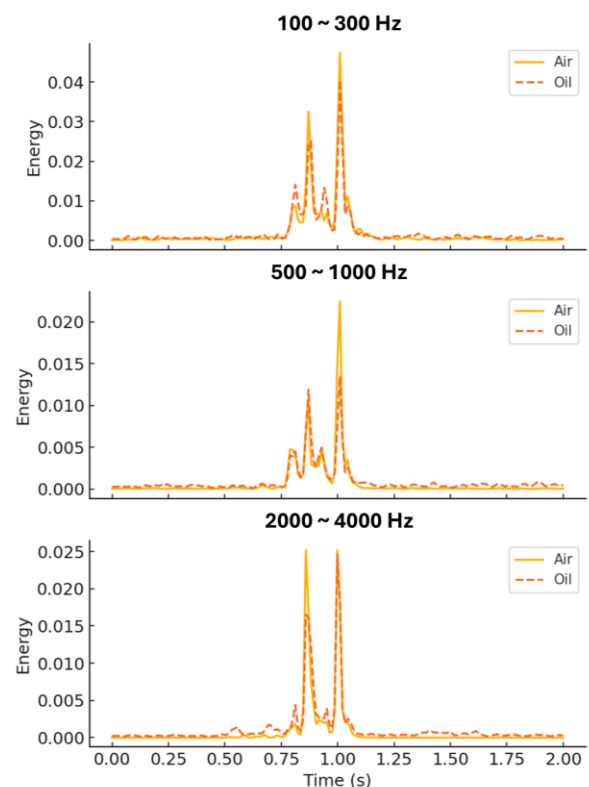


FIGURE 6. Comparison of Temporal Energy Variations across Frequency Bands (100~300 Hz, 500~1000 Hz, and 2000~4000 Hz).

frequency-domain PSDs, time-frequency STFT spectrograms, and autocorrelation characteristics.

In the time domain, initial events are characterized by a short-duration, high-amplitude impulsive peak followed by rapid decay, whereas pre-failure events exhibit increased background vibration/noise and more widely distributed amplitude fluctuations before and after the main event. This behavior suggests that, as degradation progresses, structural wear, increased backlash, or changes in contact conditions can introduce mechanical vibration components that persist beyond the nominal event interval.

Fig. 7 (left) shows a time-domain overlay of the waveforms for 240321 (initial tap-change event, yellow) and 250416 (pre-failure event, orange), and Fig. 7 (right) presents the corresponding amplitude histograms. The initial waveform exhibits a sharp pulse-like amplitude rise around 1-s, while the amplitude remains close to zero before and after the impulse. In contrast, the pre-failure waveform contains persistent low-amplitude fluctuations (noise/vibration) from the beginning, reaches its peak near 1-s, and maintains noticeable oscillations afterward, resulting in a higher overall energy level. Although the peak amplitude of the initial event (1.0) exceeds that of the pre-failure event (0.756), the RMS value increases from 0.073 (initial) to 0.114 (pre-failure), i.e., approximately a $1.6\times$ increase, indicating stronger sustained fluctuations in the pre-failure signal. The crest factor (peak-to-RMS ratio) decreases from approximately 13.8 (initial) to 6.6 (pre-failure), and the kurtosis also drops from 38.5 to 4.28, implying that the pre-failure amplitude distribution is less impulsive and more broadly spread. Overall, the initial event resembles a spike-dominated signal with a largely quiescent background, whereas the pre-failure event exhibits persistent vibration/noise with a relatively smaller but more dispersed amplitude pattern.

The Welch PSD comparison in Fig. 8 shows that both conditions are dominated by low-to-mid frequency components associated with OLTC operation, while the pre-failure condition exhibits noticeable changes in energy distribution across certain bands. In particular, the pre-failure event may show increased relative energy in low-frequency ranges or a change in spectral slope, suggesting that changes in mechanical impacts and structural resonance during degradation can be reflected in the frequency distribution.

Fig. 8 compares the Welch PSDs (dB scale) for the initial (blue) and pre-failure (red) events. The overall

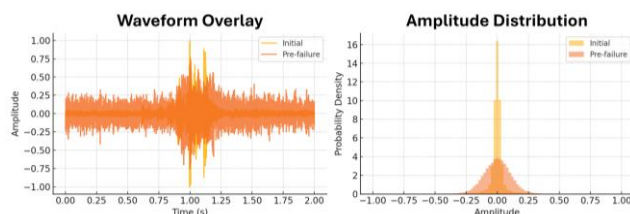


FIGURE 7. Time-domain Waveform Overlay (left) and Amplitude Distribution (right) for Initial and Pre-failure Events.

spectral shapes are broadly similar, with a log-PSD correlation coefficient of approximately 0.85, indicating that the two spectra share a comparable global structure. Both signals concentrate substantial energy in the sub-kHz range; however, the dominant peak occurs around 193 Hz for the initial event, whereas the pre-failure event peaks around 269 Hz. The initial spectrum exhibits a sharper, narrowband dominant component (-3 dB bandwidth ≈ 21 Hz), while the pre-failure dominant peak is broader with a lower Q-factor (-3 dB bandwidth ≈ 43 Hz, roughly $2\times$ wider). This suggests that the initial event produces a short, strong resonance at a specific frequency, whereas the pre-failure event disperses energy over a wider range and exhibits less distinct resonance. The spectral centroid shifts from approximately 1,393 Hz (initial) to 1,111 Hz (pre-failure), indicating that the pre-failure spectrum is, on average, biased toward lower frequencies. Consistently, 90% of the initial energy is contained within a band of approximately 4.87 kHz (e.g., roughly 0–5 kHz), whereas 90% of the pre-failure energy lies within approximately 3.40 kHz (roughly 0–3.4 kHz), implying a narrower effective bandwidth. For both signals, high-frequency content decays rapidly and becomes negligible near 11 kHz, which can be regarded as an approximate roll-off frequency. Overall, while both events have meaningful energy primarily within 0–10 kHz, the pre-failure event contains particularly less energy above ~ 5 kHz, indicating a stronger concentration of spectral energy in lower-frequency regions.

In the time-frequency domain, the initial event exhibits energy concentrated within a narrow temporal region, whereas the pre-failure event shows increased background components around the event, yielding a comparatively flatter energy distribution over the time axis. This observation suggests that compound phenomena during degradation—such as friction, arcing, and contact bounce—may affect the time-frequency pattern.

Fig. 9 presents the log-magnitude spectrograms (time-frequency distributions, dB scale) for the initial event (top) and the pre-failure event (bottom). A key difference is the presence of persistent background energy before and after the main event in the pre-failure case. The initial spectrogram shows a highly localized energy burst around

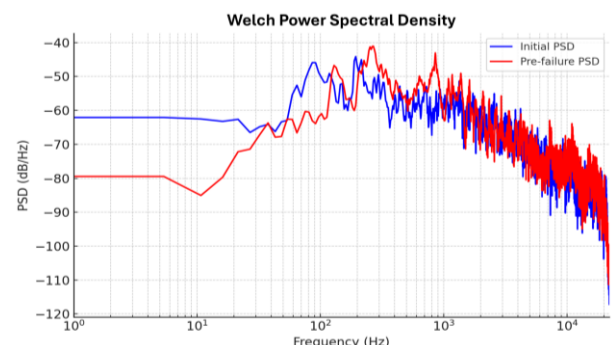


FIGURE 8. Comparison of Welch Power Spectral Density (PSD) for Initial and Pre-failure Events.

1-s, where broadband components appear simultaneously, primarily within 0~5 kHz, while the remaining time interval stays near the noise floor. In contrast, the pre-failure spectrogram exhibits noticeable energy even before the event, particularly in low-frequency bands, and the main burst near 1-s is slightly weaker but lasts longer; residual oscillations also persist after the event. When the full spectrogram matrices are vectorized and compared, the correlation coefficient is approximately 0.15, reflecting substantial differences in temporal evolution. Nevertheless, the dominant frequency-band locations at the main event instant remain similar. Overall, the initial event concentrates energy within a short interval, whereas the pre-failure event distributes energy over a broader time span with elevated background components, indicating that degradation can be diagnosed through time-frequency patterns.

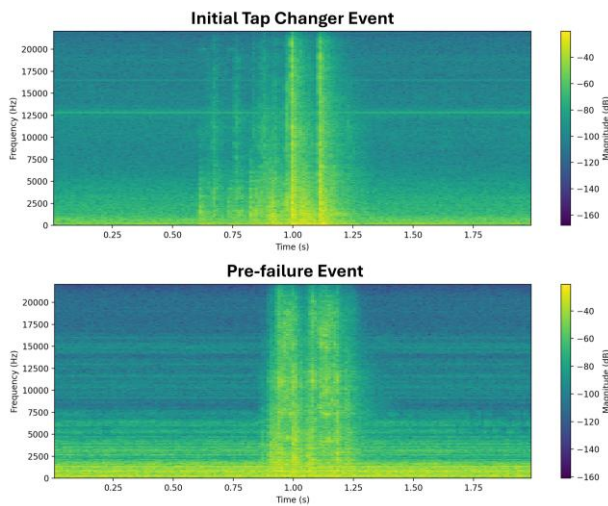


FIGURE 9. STFT Spectrogram Comparison between Initial and Pre-failure Events.

E. Motivation and Rationale for STFT-Based Frequency-Bin Prediction

The raw acoustic waveform, sampled at $f_s(\text{Hz})$, can be expressed as a one-dimensional discrete-time signal $x[n]$, where n denotes the sample index and $x[n]$ is the amplitude at the n -th sample. The relationship between the sample index n and the physical time $t(\text{s})$ is given by $t = \frac{n}{f_s}$. The number of samples contained in a segment of length $T(\text{s})$ is computed as in Equation (1).

$$L = f_s T \quad (1)$$

Here, T denotes the target prediction horizon (in seconds), f_s is the sampling frequency (Hz), and L is the total number of samples in the segment. In this work, f_s was fixed to 44,100 Hz and the event-analysis window length was set to $T=2\text{-s}$; therefore, each event consists of $L=44,100 \times 2=88,200$ samples.

If a segment of length T must be forecast repeatedly for K future steps, the total number of samples to be predicted increases as shown in Equation (2).

$$L_{\text{pred}} = f_s T K \quad (2)$$

Here, K is the number of repeated prediction steps (an integer), and L_{total} denotes the overall prediction sequence length (in samples). For example, when $K=5$, L_{total} increases to $44,100 \times 2 \times 5=441,000$. Directly forecasting a sample-level sequence of this length is computationally prohibitive and difficult to learn; in particular, self-attention-based models suffer from a bottleneck where computational complexity grows approximately as $O(L^2)$ with sequence length.

Instead of forecasting the raw waveform directly, we transform the signal using the short-time Fourier transform (STFT) to obtain a time-frequency spectrogram, and then reformulate the task as a frame-level time-series forecasting problem. STFT segments the signal using a window of length N and performs a Fourier transform on each segment to estimate time-varying spectral components. The discrete STFT is defined as in Equation (3).

$$X(m, k) = \sum_{n=0}^{N-1} x[mH + n]w[n]e^{-\frac{j2\pi kn}{N}} \quad (3)$$

In Equation (3), $x[n]$ is the input discrete signal, $w[n]$ is a window function of length N , N is the window length (in samples), H is the hop size (in samples), m is the time-frame index, k is the frequency-bin index, j is the imaginary unit, and e denotes the complex exponential. In this work, we used $N=1024$ with 50% overlap, yielding a fixed hop size of $H=512$.

For real-valued signals, we use a one-sided spectrum whose frequency axis spans 0 Hz to the Nyquist frequency $\frac{f_s}{2}$. The number of frequency-bins B is determined as in Equation (4).

$$B = \frac{N}{2} + 1 \quad (4)$$

Here, B is the number of frequency-bins. When N is even, the one-sided spectrum includes the Nyquist bin, so $B = \frac{N}{2} + 1$. With $N=1024$, $B=513$. The center frequency of the k -th bin is defined accordingly, and the frequency resolution is $\frac{f_s}{N}(\text{Hz})$.

The number of time frames M in the STFT depends on the signal length L , window length N , hop size H , and the boundary/padding policy. In our implementation, we apply zero padding at the boundaries and pad the signal at the end so that the analysis window covers both ends of the signal and the total length becomes an integer multiple of the hop size.

After boundary extension and padding, the effective lengths can be expressed as in Equation (5).

$$L_{ext} = L + N, L_{pad} = L_{ext} + \left(H - ((L_{ext} - N) \bmod H) \right) \bmod H \quad (5)$$

Here, L_{ext} is the length after adding $\frac{N}{2}$ zeros at both ends (i.e., an increase of N in total), and L_{pad} is the final length after appending zeros so that $(L_{ext} - N)$ is divisible by H . The operator mod denotes the remainder.

The final number of frames M is computed as in Equation (6).

$$M = \frac{L_{pad} - N}{H} + 1 \quad (6)$$

In our setting, the event window is $T=2$ -s, so $L=88,200$. Substituting into the above expressions yields $L_{ext}=89,224$. Since $L_{ext} - N$ leaves a remainder of 136 when divided by $H=512$, the additional padding length becomes $512 - 136 = 376$, resulting in $L_{pad}=89,600$. Consequently, $M=174$, and each event is represented as a fixed-size 513×174 time-frequency matrix.

We construct the spectrogram using the power of the complex STFT spectrum and apply a logarithmic scaling, as shown in Equation (7).

$$S_{dB}(m, k) = 10 \log_{10}(|X(m, k)|^2 + \varepsilon) \quad (7)$$

In Equation (7), S denotes the spectrogram value in dB scale, $|\cdot|$ is the complex magnitude, and ε is a small positive constant to avoid $\log(0)$; we set $\varepsilon = 10^{-6}$. This transformation compresses the dynamic range of the signal and stabilizes visualization and learning of time-frequency patterns.

Early-stage and pre-failure conditions exhibit different distributional characteristics in the time-domain waveform, PSD, and spectrogram patterns. Even for the same tap-change operation, variations in contact condition, mechanical backlash, and friction can alter when (in time) and where (in frequency) energy increases occur. Therefore, interpreting the spectrogram as a collection of frequency-bin-wise time series and forecasting its evolution is well suited for learning structured state changes.

Direct long-horizon forecasting of the sample-level waveform is impractical due to the sequence-length and computational-complexity bottleneck. In contrast, the STFT-based representation compresses each 2-s event into 174 frames, substantially reducing the temporal prediction length, while retaining 513 frequency-bins that encode state-dependent spectral energy variations. Accordingly, we formulate the task as forecasting future frame values of the spectrogram, treating each frequency-bin ($k=0 \dots 512$) as an independent (or multivariate) time series, as summarized in Fig. 10 and Table I.

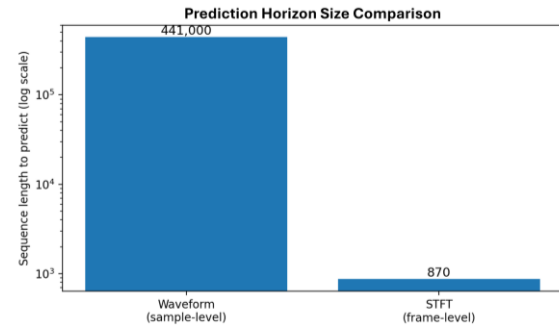


FIGURE 10. Comparison of Prediction Time-axis Lengths in the Waveform (sample-level) and STFT (frame-level) Representations (e.g., forecasting a 2-s segment for $K=5$ steps).

F. Data Preprocessing and Dataset

To extract impulsive events produced during OLTC tap-changes from the raw acoustic recordings, we use a normalized time-domain waveform $x[n]$, where n is the sample index and $x[n]$ is the amplitude at the n -th sample. After normalization, the signal range is limited to $[-1, 1]$. Event candidates are defined as local maxima of $|x[n]|$ whose amplitudes exceed a threshold θ .

The basic event-detection criterion is given in Equation (8).

$$\max_n |x[n]| \geq \theta \quad (8)$$

Here, θ is the event-detection threshold, set to $\theta=0.7$ in this study. To prevent multiple peaks within the same event from being detected redundantly, we also enforce a minimum inter-peak spacing of at least 1-s.

Let n_c denote the center-sample index of a detected event. We extract a 2-s signal snippet that includes 1-s before and 1-s after the event center, as given in Equation (9).

$$x_e[n] = x[n_0 - T_s : n_0 + T_s] \quad (9)$$

Here, $x_e[n]$ denotes the event-centered window (event snippet), and N_{1s} is the number of samples corresponding to 1-s. Using the sampling frequency f_s , N_{1s} is computed as in Equation (10).

TABLE I
DIMENSIONALITY COMPARISON BETWEEN DIRECT WAVEFORM FORECASTING AND STFT-BASED FORECASTING (2-S WINDOW, $K=5$)

Representation	Length for 2-s (step 1)	Length for 2-s $\times$ K ($K=5$)	Dim. per time step
Waveform (sample-level)	$L = 88,200$	$L_{pred} = 441,000$	1
STFT (frame-level)	$M = 174$	$M \times K = 870$	513

$$T_s = \frac{1}{f_s} \quad (10)$$

With $f_s=44,100$ Hz, $N_{1s}=44,100$ and the total snippet length becomes $2 \times N_{1s}=88,200$ samples. If the event occurs near the beginning or end of the recording and the extraction window extends beyond the signal boundaries, we pad the missing portion with zeros so that every snippet has a fixed length of 88,200 samples.

For each event snippet $x_e[n]$, we apply the STFT to generate a time-frequency spectrogram. The STFT is expressed as in Equation (11).

$$x_e(m, k) = \sum_{n=0}^{N-1} x_e[mH + n]w[n]e^{-\frac{j2\pi kn}{N}} \quad (11)$$

In Equation (11), $x_e(m, k)$ is the complex STFT spectrum of event e , $w[n]$ is a Hann window of length N , H is the hop size, m is the time-frame index, and k is the frequency-bin index. We use $N=1024$ with 50% overlap, i.e., $H=512$.

The spectrogram is obtained by applying a logarithmic scale to the STFT power and converting it to dB units, as in Equation (12).

$$S_e(m, k) = 10 \log_{10}(|x_e(m, k)|^2 + \varepsilon) \quad (12)$$

Here, $S_e(m, k)$ is the spectrogram value (dB) for event e , and ε is a small constant for numerical stability; we set $\varepsilon = 10^{-6}$. Because a one-sided spectrum is used for real-valued signals, the number of frequency-bins is fixed to $B = \frac{N}{2} + 1$.

With boundary extension and padding, the number of time frames is fixed to 174. Therefore, each event is represented as a fixed-size matrix, as in Equation (13).

$$S_e \in \mathbb{R}^{513 \times 174} \quad (13)$$

Here, $S_e \in \mathbb{R}^{513 \times 174}$ is the spectrogram matrix with 513 frequency-bins spanning 0 Hz to 22,050 Hz and 174 time frames corresponding to the 2-s event window.

To convert event-wise spectrogram matrices into a learnable long multivariate time series, we concatenate, across events, the row corresponding to the same frequency-bin k . Let e denote the event index and S_e the spectrogram of the e -th event. Then, the full time series for bin k is defined as in Equation (14).

$$f_k = [S_{1(k,:)}, S_{2(k,:)}, \dots, S_{(N_{evt})(k,:)}] \quad (14)$$

Here, f_k is a one-dimensional time series obtained by concatenating the 174-frame vectors $S_{e(k,:)}$ across events in chronological order. Since each $S_{e(k,:)}$ has length $M=174$, the total length of f_k is $M \cdot N_{evt}$, where $N_{evt}=195,994$ is the total number of tap-change events.

After constructing f_k for all frequency-bins ($k=0 \dots 512$), we stack them to form a multivariate time-series matrix X , as in Equation (15).

$$X = [f_0^T; f_1^T; \dots; f_{512}^T]^T \quad (15)$$

This representation conceptually corresponds to a matrix whose columns are frequency-bin-wise features. In practice, we store X with shape $(T_{total}, 513)$, where each row corresponds to a 513-dimensional spectral vector at a given time frame. The total number of rows is given in Equation (16).

$$N_{rows} = 174 \cdot N_{evt} \quad (16)$$

With $N_{evt}=195,994$ and 174 frames per event, the total number of rows becomes $T_{total} = 174 \times 195,994 = 34,102,956$.

We split the dataset at the event level into Train/Validation/Test subsets with a 60/20/20 ratio. The resulting event counts are summarized in Table II. The corresponding number of frames (rows) is computed as (number of events) $\times$ 174. Model inputs are formed by sampling contiguous segments from X using a sliding-window strategy.

IV. METHOD

In this section, we first define the core components of the baseline model, MoEKAN, and then describe how the proposed model, AcousticMoEKAN, extends MoEKAN through architectural modifications. We assume that a preprocessed multivariate time-series input (e.g., from STFT) is already available, and we focus on input/output dimensions, gating, mathematical definitions of experts, and expert-combination rules.

AcousticMoEKAN adopts a KAN-based Mixture-of-Experts inductive bias tailored to the time-frequency evolution of OLTC tap-changing acoustics: after mapping each event to an STFT magnitude spectrogram, the per-frequency-bin trajectories combine short impulsive excitations, band-limited resonances, envelope-like energy bursts, and slower drifts, so robust long-horizon forecasting benefits from decomposing the function space into a small set of pattern-aligned components rather than relying on a

TABLE II
EVENT-WISE DATA SPLIT AND CORRESPONDING NUMBER OF FRAMES
(174 FRAMES/EVENT)

Split	No. of events	No. of frames (rows)	Ratio
Train	117,596	20,461,704	60%
Validation	39,199	6,820,626	20%
Test	39,199	6,820,626	20%
Total	195,994	34,102,956	100%

single uniform parametrization. Each expert is implemented as a KAN, where each connection learns a one-dimensional nonlinear function, enabling explicit and data-efficient approximation of diverse nonlinear shapes; the expert pool is constructed to span dominant acoustic dynamics—Taylor and Linear experts for smooth trend/drift, Wavelet and B-spline experts for localized transients and sharp deviations, Fourier experts for oscillatory structures, Jacobi experts for stable orthogonal approximation of smooth nonlinear patterns, ExpDecay experts for post-impact ring-down/decay, GaussEnv experts for envelope-modulated energy bursts, AR experts for persistent autocorrelated behavior and structured noise-floor variation, and Gammatone experts for band-pass/resonant responses—while a sparse MoBA gate partitions the input window into temporal blocks, selects top-k salient blocks, and routes to top-k experts so that only the most informative segments and a small subset of specialists are activated, encouraging specialization, improving extrapolation under regime changes, and reducing unnecessary computation.

A. Base Model: MoEKAN

Given a multivariate time series $\{x_t\}$, we use a past window of length L to predict a future window of length H . The time index t is an integer, and the observation at each time step is represented as a D -dimensional vector.

$$x_t = [x_{t,1}, x_{t,2}, \dots, x_{t,D}]^T \in \mathbb{R}^D \quad (17)$$

In Equation (17), $x_{t,D}$ denotes the scalar observation of the d -th variable (or channel) at time t . Thus, $x_t \in \mathbb{R}^D$ stacks D observations into a single column vector, and $[\cdot]^T$ denotes transpose.

$$X_t = [x_{t-L+1}, x_{t-L+2}, \dots, x_t]^T \in \mathbb{R}^{L \times D} \quad (18)$$

In Equation (18), $X_t \in \mathbb{R}^{L \times D}$ denotes the input window provided to the model at prediction time t . The row dimension corresponds to time (L frames) and the column dimension corresponds to variables/channels (D).

$$Y_t = [x_{t+1}, x_{t+2}, \dots, x_{t+H}]^T \in \mathbb{R}^{H \times D} \quad (19)$$

In Equation (19), $Y_t \in \mathbb{R}^{H \times D}$ denotes the ground-truth window for the next H frames.

$$\hat{Y}_t = f_\theta(X_t) \quad (20)$$

In Equation (20), $f_\theta(\cdot)$ denotes a forecasting model with learnable parameters θ , and $\hat{Y}_t$ is the predicted future window output by the model.

Because the measurement environment can induce window-dependent shifts in mean and variance, MoEKAN applies RevIN (Reversible Instance Normalization). Specifically, for each input window, it computes statistics

to normalize the input, and then restores the prediction using the same statistics.

$$\mu_d = \left(\frac{1}{L}\right) \sum_{i=1}^L x_{t-L+i,d} \quad (21)$$

In Equation (21), μ_d is the mean of the d -th variable within the input window X_t . Here, i indexes the samples within the window ($1 \dots L$).

$$\sigma_d = \sqrt{\left(\frac{1}{L}\right) \sum_{i=1}^L (x_{t-L+i,d} - \mu_d)^2 + \varepsilon} \quad (22)$$

In Equation (22), σ_d is the standard deviation of the d -th variable, and $\varepsilon > 0$ is a stabilization constant to avoid division by values close to zero.

$$z_{i,d} = \frac{\gamma_d(x_{t-L+i,d} - \mu_d)}{\sigma_d} + \beta_d, \quad 1 \leq i \leq L \quad (23)$$

In Equation (23), γ_d and β_d are learnable scale and shift parameters for variable d . Let $Z_t \in \mathbb{R}^{L \times D}$ denote the RevIN-normalized input window whose (i,d) -th element is $z_{i,d}$.

$$\hat{x}_{t+j,d} = \frac{\sigma_d(\hat{z}_{j,d} - \beta_d)}{\gamma_d} + \mu_d, \quad 1 \leq j \leq H \quad (24)$$

Equation (24) performs the inverse transformation.

MoEKAN adopts a Mixture-of-Experts (MoE) architecture. Each of the E experts generates its own prediction, and a gate produces input-dependent mixing coefficients π to form the final prediction as a weighted sum.

$$\hat{Y}_t^e = f_{e(Z_t)}, \quad e = 1, \dots, E \quad (25)$$

In Equation (25), $\hat{Y}_t^e$ is the prediction produced by the e -th expert. $\hat{Y}_t$ is the prediction produced by the e -th expert given Z_t .

$$\pi = \text{softmax}(a), \quad a_e = w_e^T h + b_e \quad (26)$$

In Equation (26), h is a summary representation (embedding) extracted from the input time series by the gate, w_e^T and b_e are the gate parameters for expert e , and a_e is the resulting logit. The softmax function normalizes logits into a probability distribution π such that $\pi_e \geq 0$.

$$\hat{Y}_t = \sum_{e=1}^E \pi_e \hat{Y}_t^e \quad (27)$$

Equation (27) defines the MoE aggregation rule: the larger π_e is, the more the e -th expert contributes to the final output.

KAN (Kolmogorov-Arnold Network) replaces fixed activation functions with learnable one-dimensional functions $\varphi(\cdot)$ on each connection (edge), enabling the network to adapt its nonlinear transformations to the data. In MoEKAN, each expert is constructed using a distinct set of basis functions.

$$\varphi^{(e)}(u) = \sum_{r=1}^{R_e} c_r^e \psi_r^{(e)}(u) \quad (28)$$

In Equation (28), u is a scalar input, $\varphi^{(e)}(u)$ is the u -th basis function used by expert e , c_r^e are learnable coefficients, and R is the number of basis functions.

MoEKAN employs four experts: (i) Taylor (polynomial), (ii) Wavelet (localized), (iii) Jacobi (orthogonal polynomial), and (iv) Fourier (periodic/harmonic).

$$\psi_r^{Taylor}(u) = u^r, \quad r = 0, 1, \dots, P \quad (29)$$

In Equation (29), P is the maximum order. Polynomial bases are suitable for modeling smooth trends and gradual curvature.

$$\psi_{s,\tau}^{Wavelet}(u) = \left(\frac{1}{\sqrt{s}}\right) \psi\left(\frac{u-\tau}{s}\right) \quad (30)$$

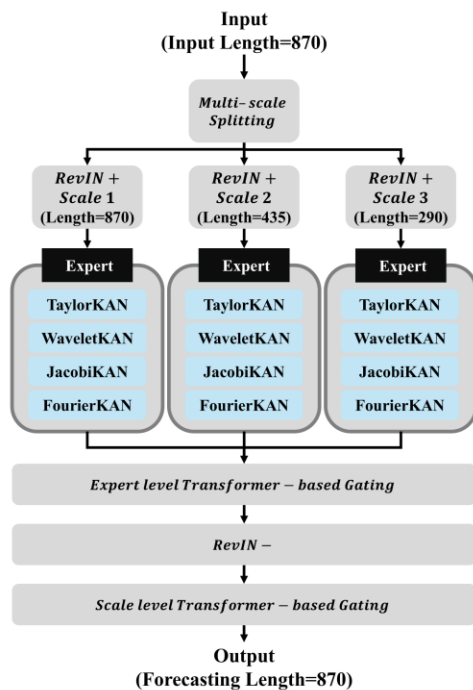


FIGURE 11. The Structure of MoEKAN.

In Equation (30), s is the scale and τ is the translation parameter. Wavelet bases are sensitive to time-localized changes.

$$\psi_r^{Jacobi}(u) = P_r^{(\alpha,\beta)}(u) \quad (31)$$

In Equation (31), α and β control the basis shape. Orthogonal-polynomial approximation is numerically stable.

$$\psi_r^{Fourier}(u) = \sin(r \omega u) \text{ or } \cos(r \omega u) \quad (32)$$

In Equation (32), ω is the fundamental angular frequency and r is the harmonic index. Fourier bases are suitable for representing periodic and harmonic components.

MoEKAN employs a self-attention-based Transformer encoder as the gate. It forms contextual representations from input tokens and uses them to compute expert logits.

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (33)$$

In Equation (33), Q , K , and V denote the query/key/value matrices, and d_k is the key-vector dimension. Dense gating computes continuous mixing coefficients π over all experts.

$$\mathcal{L}_{MSE} = \left(\frac{1}{HD}\right) \|Y_t - \hat{Y}_t\|_F^2 \quad (34)$$

In Equation (34), $\|\cdot\|_F$ denotes the Frobenius norm, corresponding to the sum of squared errors over all elements. Thus, the loss is the mean squared error normalized over the prediction horizon H and the number of variables D .

B. Proposed Model: AcousticMoEKAN

AcousticMoEKAN preserves the forecasting framework of MoEKAN but (i) expands the expert set from 4 to 10 to enlarge the function space and (ii) replaces the dense Transformer gate with a sparse Mixture-of-Block Attention (MoBA) gate so that only a small subset of experts is activated in an input-adaptive manner.

The input/output definitions, RevIN normalization, and MoE aggregation rule of AcousticMoEKAN follow MoEKAN. The key differences lie in the expert set (number and composition) and in the gating implementation, which we detail below.

$$E_{MoEKAN} = 4, \quad E_{AcousticMoEKAN} = 10 = 4 + 6 \quad (35)$$

In Equation (35), E denotes the number of experts. AcousticMoEKAN retains the original four experts (Taylor/Wavelet/Jacobi/Fourier) and introduces six

additional experts to broaden the representable pattern space.

$$\psi^{Linear(u)} = au + b \quad (36)$$

In Equation (36), a and b are the slope and intercept, respectively, enabling direct approximation of linear trends and drift.

$$\psi^{ExpDecay(u)} = \exp(-\lambda u) \quad (37)$$

In Equation (37), $\lambda > 0$ is a decay rate, modeling exponentially decaying patterns such as post-impact reverberation.

$$\psi^{GaussEnv(u)} = \exp\left(-\frac{(u - \mu_g)^2}{2\sigma_g^2}\right) \quad (38)$$

In Equation (38), μ is the center location and σ is the width, capturing concentration and dispersion in an energy envelope.

$$y_t = \sum_{i=1}^p \varphi_i y_{t-i} + \varepsilon_t \quad (39)$$

Equation (39) is a p -th order autoregressive (AR) form. y_t denotes the value at time t , φ_i are regression coefficients, and ε_t is a residual term, directly reflecting strong autocorrelation structures.

$$g(t) = At^{n-1} \exp(-2\pi bt) \cos(2\pi f_c t + \phi) \quad (40)$$

Equation (40) provides an example of a resonant response in gammatone form. A is a scale factor, n is the order, b controls decay/bandwidth, f_c is the center frequency, and ϕ is the phase.

$$B_i^p = \alpha_i^p B_i^{p-1} + \beta_i^p B_{i+1}^{p-1} \quad (41)$$

Equation (41) provides a compact notation for the recursive (Cox-de Boor) construction of B-spline bases.

$$\hat{\mathbf{Y}}_t^{(original)} = \text{RevIN}^{-1}(\hat{\mathbf{Y}}_t)$$

The MoBA gate combines (i) temporal block partitioning, (ii) top-k selection of salient blocks, (iii) construction of a summary representation using only the selected blocks, and (iv) sparse routing that activates only the top-k experts [51].

$$I_b = \{(b-1)\ell + 1, \dots, b\ell\}, \quad b = 1, \dots, B \quad (42)$$

In Equation (42), ℓ is the block length (in frames), B is the number of blocks, and I_b denotes the index set of the b -th block.

$$b_b = \left(\frac{1}{\ell}\right) \sum_{t \in I_b} h_t \quad (43)$$

In Equation (43), h_t denotes a token representation.

$$s_b = v^T b_b, \quad S = \text{TopK}(\{s_b\}_{b=1}^B, k_B) \quad (44)$$

In Equation (44), v is a learnable parameter vector, s_b is the block-importance score, and S is the index set of the top-k blocks.

$$\hat{h} = \text{Pool}(\{h_t | t \in \cup_{b \in S} I_b\}) \quad (45)$$

In Equation (45), h is the summary representation computed using only the selected blocks; Pool can be implemented as average pooling or restricted attention, among other choices.

$$a_e = w_e^T \hat{h} + b_e, \quad K = \text{TopK}(\{a_e\}_{e=1}^E, k_E) \quad (46)$$

In Equation (46), a_e is the expert logit and K is the index set of the top-k experts.

$$\pi_e = \frac{1_{e \in K} \exp(a_e)}{\sum_{j \in K} \exp(a_j)} \quad (47)$$

Equation (47) defines sparse softmax (keep-top-k) mixing coefficients.

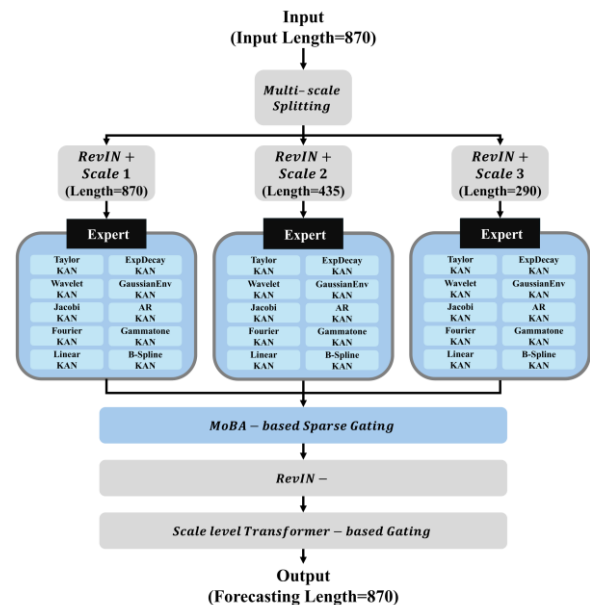


FIGURE 12. The Structure of AcousticMoEKAN.



FIGURE 13. Comparison of Transformer Dense Gating and MoBA Sparse Gating.

TABLE III
SUMMARY OF ARCHITECTURAL DIFFERENCES
BETWEEN MOEKAN AND ACOUSTICMOEKAN

Component	MoEKAN (Base)	AcousticMoEKAN (Proposed)
No. of experts E	4	10 (4 + 6)
Base experts	Taylor, Wavelet, Jacobi, Fourier	Same
Additional experts	None	Linear, ExpDecay, GaussEnv, AR, Gammatone, B-spline
Gating	Transformer + dense softmax	MoBA: block top-k + expert top-k (sparse)
Mixture weights π	All experts active	Only selected K active; others set to 0
Objective	$\mathcal{L}_{MSE}$	$\mathcal{L}_{MSE} + \text{load-balancing}$

$$\mathcal{L} = \mathcal{L}_{MSE} + \lambda_{lb} \sum_{e=1}^E \left(\rho_e - \frac{1}{E} \right)^2 \quad (48)$$

Equation (48) augments the objective with a load-balancing auxiliary loss to encourage balanced expert utilization. ρ_e is the fraction of times expert e is selected within a batch, λ_{lb} is the auxiliary-loss weight.

Table III summarizes the key differences between the two models. By expanding the expert pool and introducing sparse gating, AcousticMoEKAN enhances both representational capacity and selective computation.

V. RESULTS AND DISCUSSION

In this section, we quantitatively and qualitatively compare the proposed AcousticMoEKAN with a diverse set of time-series forecasting baselines spanning linear, CNN, GNN, Transformer, and KAN-based models. All models are trained under the same data split (train/validation/test) and the same forecasting protocol (input length L and prediction horizon H). We report mean squared error (MSE), mean absolute error (MAE), and mean absolute percentage error (MAPE) on the test set. In addition, to examine frequency-dependent forecasting difficulty, we partition the STFT spectrogram frequency axis into five regions (including the DC component) and analyze model behavior across bands.

The input to all models is an STFT-based spectrogram time series, where the number of frequency-bins is set to $K=513$ (0 Hz~22.05 kHz). The center frequency of the k -th bin is determined by the sampling frequency f_s and the STFT window length N ($=1024$) as shown in Equation (49).

$$f_k = k \cdot \frac{f_s}{N}, \quad k = 0, 1, \dots, K-1 \quad (49)$$

In Equation (49), f_k denotes the center frequency (Hz) of the k -th bin, f_s is the sampling frequency (44.1 kHz in this work), and N is the STFT window length; the frequency resolution between adjacent bins is $\Delta f = \frac{f_s}{N}$. To examine the frequency dependence of forecasting performance, we divide the frequency axis into five regions. Specifically, (1) $k=0$ corresponds to the 0 Hz (DC) component; (2) $k=1\sim128$ corresponds to 43.07~5512.50 Hz; (3) $k=129\sim256$ corresponds to 5555.57~11025.00 Hz; (4) $k=257\sim384$ corresponds to 11068.07~16537.50 Hz; and (5) $k=385\sim512$ corresponds to 16580.57~22050.00 Hz. For each region, the reported value is computed by averaging the errors over the bins belonging to that region.

Prediction errors between model outputs and ground truth are computed on the time–frequency grid. Let T denote the prediction horizon (or, equivalently, index the evaluated time steps as $t=1, \dots, T$ across all test samples). The MSE for band r is defined in Equation (50) as follows.

$$MSE_r = \left(\frac{1}{|B_r| \cdot T} \right) \cdot \sum_{k \in B_r} \sum_{t=1}^T (y_{\{k,t\}} - \hat{y}_{\{k,t\}})^2 \quad (50)$$

TABLE IV

FREQUENCY-BAND-WISE FORECASTING ERRORS (MSE/MAE/MAPE) AND OVERALL AVERAGES FOR ALL COMPARED MODELS

Type	Network	Frequency (Hz)	MSE	MAE	MAPE	AVERAGE MSE	AVERAGE MAE	AVERAGE MAPE
Linear	NLinear	0.00	144.83	8.96	15.72	99.31	7.66	11.42
		43.07 ~ 5512.50	68.01	6.79	14.47			
		5555.57 ~ 11025.00	114.54	8.52	10.14			
		11068.07 ~ 16537.50	98.71	7.49	9.72			
		16580.57 ~ 22050.00	70.46	6.54	7.04			
	RLinear	0.00	144.99	9.01	15.95	80.15	6.63	9.82
		43.07 ~ 5512.50	33.41	4.35	9.25			
		5555.57 ~ 11025.00	81.19	7.09	8.57			
		11068.07 ~ 16537.50	80.41	6.71	8.76			
		16580.57 ~ 22050.00	60.73	5.97	6.55			
GNN	STID	0.00	137.96	8.33	14.19	80.43	6.31	9.35
		43.07 ~ 5512.50	35.41	4.25	9.01			
		5555.57 ~ 11025.00	70.57	6.28	7.87			
		11068.07 ~ 16537.50	105.08	7.29	9.82			
		16580.57 ~ 22050.00	53.12	5.40	5.85			
	TimeGNN	0.00	361.80	15.74	30.00	386.93	13.73	21.90
		43.07 ~ 5512.50	85.87	6.93	16.03			
		5555.57 ~ 11025.00	514.77	17.68	23.76			
		11068.07 ~ 16537.50	683.05	17.72	26.32			
		16580.57 ~ 22050.00	289.17	10.60	13.38			
CNN	SCINet	0.00	252.92	11.99	22.64	260.05	12.07	16.19
		43.07 ~ 5512.50	50.46	5.53	12.31			
		5555.57 ~ 11025.00	603.27	21.83	22.05			
		11068.07 ~ 16537.50	258.42	12.89	15.08			
		16580.57 ~ 22050.00	135.20	8.13	8.88			
	TimesNet	0.00	122.22	7.85	13.57	172.29	8.44	12.77
		43.07 ~ 5512.50	58.66	6.15	13.81			
		5555.57 ~ 11025.00	109.30	7.45	8.44			
		11068.07 ~ 16537.50	510.91	15.28	22.12			
		16580.57 ~ 22050.00	60.35	5.48	5.89			
Transformer	PatchTST	0.00	142.33	8.76	16.07	98.22	7.10	10.72
		43.07 ~ 5512.50	37.72	4.44	9.61			
		5555.57 ~ 11025.00	137.22	8.64	11.42			
		11068.07 ~ 16537.50	101.00	7.24	9.60			
		16580.57 ~ 22050.00	72.84	6.43	6.88			
	iTransformer	0.00	170.67	9.72	18.72	101.45	7.33	10.79
		43.07 ~ 5512.50	39.29	4.94	10.43			
		5555.57 ~ 11025.00	145.43	9.52	9.95			
		11068.07 ~ 16537.50	105.79	7.44	9.41			
		16580.57 ~ 22050.00	46.09	5.06	5.45			
KAN	DenseRMoK	0.00	114.47	8.08	14.76	118.54	7.53	11.58
		43.07 ~ 5512.50	33.52	4.29	9.37			
		5555.57 ~ 11025.00	236.28	10.94	15.22			
		11068.07 ~ 16537.50	139.06	7.99	11.51			
		16580.57 ~ 22050.00	69.38	6.32	7.02			
	MoEKAN	0.00	128.94	8.18	14.50	75.66	6.21	9.31
		43.07 ~ 5512.50	33.53	4.14	8.95			
		5555.57 ~ 11025.00	78.88	6.59	8.44			
		11068.07 ~ 16537.50	77.52	6.44	8.38			
		16580.57 ~ 22050.00	59.43	5.68	6.29			
	Acoustic MoEKAN	0.00	110.57	7.57	13.07	60.06	5.62	8.23
		43.07 ~ 5512.50	30.56	4.17	8.90			
		5555.57 ~ 11025.00	58.09	5.75	6.54			
		11068.07 ~ 16537.50	57.88	5.58	7.28			
		16580.57 ~ 22050.00	43.20	5.02	5.34			

The MAE is defined in Equation (51) as follows.

$$MAE_r = \left(\frac{1}{|B_r| \cdot T} \right) \cdot \sum_{\{k \in B_r\}} \sum_{\{t=1\}}^{\{T\}} |y_{\{k,t\}} - \hat{y}_{\{k,t\}}| \quad (51)$$

In Equation (50) and (51), $y_{\{k,t\}}$ denotes the ground-truth value (e.g., a log-magnitude spectrum in dB) at time index t and frequency-bin k , $\hat{y}_{\{k,t\}}$ is the corresponding prediction. MAPE reflects relative error with respect to the target magnitude; to avoid instability when the target is 0 or close to 0, we compute MAPE with a small constant ε as in Equation (52).

$$MAPE_r = \left(\frac{100}{|B_r| \cdot T} \right) \cdot \sum_{\{k \in B_r\}} \sum_{\{t=1\}}^{\{T\}} \frac{|y_{\{k,t\}} - \hat{y}_{\{k,t\}}|}{|y_{\{k,t\}}| + \varepsilon} \quad (52)$$

Finally, the overall performance reported as “AVERAGE” is defined as the equally weighted mean of the five band-wise scores.

Note that the DC component (0 Hz) comprises a single bin. Because our tables compute “AVERAGE” by equally weighting the five regions, the DC band can have a disproportionately large influence on the overall average relative to its bin count. Therefore, we jointly report (i) band-wise performance and (ii) the overall average, enabling simultaneous interpretation of frequency-dependent difficulty and model-specific strengths/weaknesses.

We benchmark against 11 models that represent major architectural families. The linear baselines are NLinear and RLinear; the GNN baselines are STID and TimeGNN; the CNN baselines are SCINet and TimesNet; the Transformer baselines are PatchTST and iTransformer. The KAN family includes DenseRMoK, MoEKAN, and the proposed AcousticMoEKAN. This suite is designed to position the proposed method against: (i) strong linear mappings, (ii) graph-based relational learning, (iii) convolutional local-pattern modeling, (iv) attention-based long-range dependency modeling, and (v) KAN-based adaptive nonlinear approximation.

Table IV reports frequency-band-wise forecasting errors (MSE/MAE/MAPE) for all compared models, together with the overall average (AVERAGE). The frequency axis consists of 513 STFT bins ($k=0, \dots, 512$), where the $k=0$ (0 Hz, DC) component is reported as a separate region. The remaining 512 bins ($k=1 \sim 512$) are evenly divided into four contiguous bands of 128 bins each ($k=1 \sim 128$, $129 \sim 256$, $257 \sim 384$, and $385 \sim 512$), and the mean error is computed per band. The “AVERAGE” value is the equally weighted mean over the five regions including DC, serving as a summary metric for overall performance without bias toward a particular frequency region.

Table V summarizes the overall performance of all 11 models using the equally weighted average over the five frequency bands (AVG). AcousticMoEKAN ranks first across all metrics, achieving an average MSE of 60.06, MAE of 5.62, and MAPE of 8.23. Relative to the base MoEKAN, AcousticMoEKAN reduces MSE by approximately 20.6%, MAE by 9.5%, and MAPE by 11.6%, confirming a systematic structural improvement for spectrogram time-series forecasting.

On the overall average, KAN-based models form the leading performance tier. AcousticMoEKAN and MoEKAN occupy the top two positions, followed by RLinear (AVG MSE=80.15) and STID (AVG MSE=80.43) with a small margin. In contrast, TimeGNN exhibits substantially larger errors across all bands (AVG MSE=386.93), suggesting that explicit graph-structure learning is not effective under this dataset configuration. The CNN-based models (SCINet and TimesNet) remain competitive in certain bands but rank in the middle-to-lower tier in terms of the overall average.

From a metric perspective, MSE assigns a quadratic penalty to large deviations (outliers), whereas MAE captures the typical error magnitude, and MAPE emphasizes relative error normalized by the target magnitude. Consequently, a model that is generally accurate but occasionally produces large mistakes may yield low MAE but high MSE, while MAPE can fluctuate

TABLE V
OVERALL AVERAGE FORECASTING PERFORMANCE ACROSS FIVE FREQUENCY BANDS

Type	Model	AVG MSE ↓	AVG MAE ↓	AVG MAPE ↓	Rank(MSE)	Rank(MAE)	Rank(MAPE)
KAN	AcousticMoEKAN	60.06	5.62	8.23	1	1	1
KAN	MoEKAN	75.66	6.21	9.31	2	2	2
Linear	RLinear	80.15	6.63	9.82	3	4	4
GNN	STID	80.43	6.31	9.35	4	3	3
Transformer	PatchTST	98.22	7.10	10.72	5	5	5
Linear	NLinear	99.31	7.66	11.42	6	8	7
Transformer	iTransformer	101.45	7.33	10.79	7	6	6
KAN	DenseRMoK	118.54	7.53	11.58	8	7	8
CNN	TimesNet	172.29	8.44	12.77	9	9	9
CNN	SCINet	260.05	12.07	16.19	10	10	10
GNN	TimeGNN	386.93	13.73	21.90	11	11	11

more strongly in regions where the target values are small. These characteristics become more evident in the band-wise analysis below.

In addition to forecasting accuracy, practical deployment requires understanding the computational and memory footprint of each model, as summarized in Table VI. To provide a coarse but informative reference, we summarize parameter counts and heuristic estimates of peak training memory, per-event FLOPs, and forward latency under a representative setting (batch size = 2; a 174-frame STFT sequence per sample; single-GPU execution). Parameter counts are taken from training logs. The peak training memory of AcousticMoEKAN (approximately 20.00 GB on an RTX 3090 with 24 GB memory) is empirically observed, whereas the remaining memory/FLOPs/latency values for other models are estimated using simple scaling rules for sanity-checking. Therefore, these numbers should not be interpreted as profiler-measured results; depending on implementation details (e.g., attention kernels, mixed precision, and activation checkpointing), the absolute error can exceed $\pm 50\%$, especially for attention- and MoE-style architectures.

The estimated footprint reveals a clear separation between lightweight baselines and higher-capacity models. Linear and ID-embedding-based approaches remain extremely compact and fast (e.g., NLinear/RLinear at approximately 0.76M parameters with about 1.07 GB peak training memory and ~ 0.28 ms per event; STID at approximately 0.11M parameters with ~ 0.84 GB and ~ 0.26 ms per event). CNN baselines are also relatively lightweight (e.g., SCINet at ~ 0.72 M parameters and TimesNet at ~ 2.00 M parameters). In contrast, models that explicitly target richer nonlinear dynamics and cross-variable interactions incur increased cost, including TimeGNN (~ 7.90 M parameters; ~ 3.65 GB; ~ 0.73 ms per event), iTransformer (~ 4.00 M; ~ 2.24 GB; ~ 0.63 ms per event), DenseRMoK (~ 10.64 M; ~ 4.63 GB; ~ 1.40 ms per

event), and MoEKAN (~ 15.67 M; ~ 6.05 GB; ~ 1.74 ms per event).

At the upper end, PatchTST (~ 51.31 M parameters) and AcousticMoEKAN (~ 53.23 M parameters) form the highest-cost tier, with estimated peak training memory of ~ 19.31 GB and ~ 20.00 GB, respectively. Under the same heuristic setting, PatchTST and AcousticMoEKAN correspond to approximately 35.71 and 44.43 G FLOPs per event, yielding roughly 4.71 and 5.80 ms per-event forward latency when assuming an effective throughput of 8 TFLOPs and a fixed 0.25 ms overhead. Relative to the base MoEKAN (15.67M parameters; ~ 6.05 GB peak training memory; 13.33 G FLOPs; ~ 1.74 ms per event), the proposed AcousticMoEKAN increases the footprint due to expert expansion and routing, but remains feasible within a single 24-GB GPU training setup. With these efficiency considerations in mind, we next interpret the results from a metric perspective and analyze frequency-dependent forecasting difficulty across bands.

Table VII summarizes, for each frequency band, the best-performing model under each metric. AcousticMoEKAN attains the lowest MSE and MAPE in all bands and also achieves the lowest MAE in most bands. The only exception is the low-frequency band 43.07–5512.50 Hz, where MoEKAN achieves a slightly lower MAE (4.14) than AcousticMoEKAN (4.17).

Table VIII summarizes the best-performing non-KAN baseline for each frequency band. The strongest non-KAN baseline varies across bands. At DC (0 Hz), TimesNet performs best; in the low-frequency band (43.07–5512.50 Hz), RLinear is best in terms of MSE while STID is best in MAE/MAPE; in the mid-frequency band (5555.57–11025.00 Hz), STID is most competitive; and in the high-frequency band (16580.57–22050.00 Hz), iTransformer achieves the best results. This variability indicates that the statistical properties of the data and the most effective inductive bias differ across frequency regions.

TABLE VI
MODEL SIZE AND COMPUTATIONAL/MEMORY FOOTPRINT OF COMPARED MODELS

Type	Model	Params (M)	Train Peak Memory (GB)	FLOPs (G/event)	Latency (ms/event)
Linear	NLinear	0.76	1.07	0.26	0.28
Linear	RLinear	0.76	1.07	0.26	0.28
GNN	STID	0.11	0.84	0.05	0.26
GNN	TimeGNN	7.90	3.65	3.85	0.73
CNN	SCINet	0.72	1.06	0.40	0.30
CNN	TimesNet	2.00	1.52	1.25	0.41
Transformer	PatchTST	51.31	19.31	35.71	4.71
Transformer	iTransformer	4.00	2.24	3.06	0.63
KAN	DenseRMoK	10.64	4.63	9.22	1.40
KAN	MoEKAN	15.67	6.05	13.33	1.74
KAN	Acoustic MoEKAN	53.23	20.00	44.43	5.80

When band difficulty is assessed by averaging errors across all models, the 43.07~5512.50 Hz band is the easiest to predict, with an average MSE of 46.04. In contrast, errors increase sharply in 5555.57~11025.00 Hz (average MSE 195.41) and 11068.07~16537.50 Hz (average MSE 201.62), indicating substantially higher forecasting difficulty. The DC component (0 Hz) also exhibits a relatively large average MSE of 166.52, which may be attributed to stronger non-stationarity—e.g., variations in overall signal energy/bias caused by changes in background noise levels or sensor offsets.

The gains of AcousticMoEKAN are particularly pronounced in the mid-to-high frequency bands. In terms of MSE, compared with MoEKAN, AcousticMoEKAN reduces error by approximately 26.4% in 5555.57~11025.00 Hz, 25.3% in 11068.07~16537.50 Hz, and 27.3% in 16580.57~22050.00 Hz. This suggests that the proposed model more stably approximates complex nonlinear fluctuations in the mid/high-frequency spectrum (e.g., broadband components induced by impacts, friction, and arcing).

In the low-frequency band 43.07~5512.50 Hz, MAE is marginally lower for MoEKAN (4.14) than for AcousticMoEKAN (4.17); however, in the same band, AcousticMoEKAN yields a lower MSE (30.56 vs. 33.53). This pattern indicates that while the typical absolute error is nearly identical, MoEKAN may occasionally incur larger deviations that are amplified under the squared-error metric. Accordingly, AcousticMoEKAN can be interpreted as providing a more stable forecast by reducing outlier errors even in the low-frequency region.

The OLTC acoustic data exhibit a mixture of abrupt energy changes driven by tap-changing events (localized spikes, decay, and reverberation) and slowly varying background dynamics along the time axis. Along the frequency axis, the spectrum is characterized by low-frequency energy dominance (mechanical impacts/drive sounds), more complex broadband components in the mid-frequency region, and reduced SNR at high frequencies (noise floor). As a result, each model's inductive bias—

linearity, locality, global dependency modeling, nonlinear approximation mechanisms, and routing strategies—translates directly into band-wise performance differences.

Fig. 14 visualizes qualitative forecasts on the test set for a representative sample segment at several frequency-bins (0.00 Hz (DC), 43.07 Hz, 5555.57 Hz, 11068.07 Hz, and 16580.57 Hz). Each row corresponds to a model and each column to a frequency-bin; blue denotes the ground truth and orange denotes the prediction. This visualization highlights (i) temporal alignment of event peaks (phase alignment), (ii) peak amplitude and sharpness, and (iii) baseline/noise-floor estimation between peaks, providing an intuitive explanation for cases where MAE and MSE induce different rankings.

At DC (0.00 Hz) and 43.07 Hz, the signals typically comprise smooth low-frequency variations with intermittent sharp dips/spikes. Linear models such as NLinear and RLinear track the overall low-frequency trend reasonably well, but they tend to average out fine-scale dynamics, leading to oversmoothing of dips in both magnitude and shape. In such cases, MAE may remain relatively small, whereas MSE can increase substantially when large errors occur around dips. In contrast, MoEKAN and AcousticMoEKAN better preserve the location and magnitude of these dips, which directly contributes to improved MSE even in the low-frequency regime.

In the mid-to-high frequency range (≥ 5555.57 Hz), repeated impulse-like peaks associated with tap-changing become prominent, and forecasting accuracy is largely governed by (a) temporal misalignment of peaks, (b) under/over-estimation of peak amplitudes, and (c) drift in the inter-peak floor. TimeGNN generally fails to reconstruct peaks and produces overly flat outputs, resulting in large errors. SCINet and TimesNet capture certain periodic aspects but can still suffer from amplitude distortion or over/under-estimation of reverberation components. Transformer-based models (PatchTST and iTransformer) often align peak timing comparatively well, yet may exhibit a mild bias in peak height and between-peak regions; consequently, their ranks can vary depending on the band and metric (MSE/MAE/MAPE).

To clarify the contribution of the key architectural design factors beyond the band-wise ranking discussion,

TABLE VII

BEST-PERFORMING MODEL FOR EACH FREQUENCY BAND AND METRIC

Frequency band (Hz)	Best MSE (↓)	Best MAE (↓)	Best MAPE (↓)
0	Acoustic MoEKAN (110.57)	Acoustic MoEKAN (7.57)	Acoustic MoEKAN (13.07)
43.07~5512.50	Acoustic MoEKAN (30.56)	MoEKAN (4.14)	Acoustic MoEKAN (8.90)
5555.57~11025.00	Acoustic MoEKAN (58.09)	Acoustic MoEKAN (5.75)	Acoustic MoEKAN (6.54)
11068.07~16537.50	Acoustic MoEKAN (57.88)	Acoustic MoEKAN (5.58)	Acoustic MoEKAN (7.28)
16580.57~22050.00	Acoustic MoEKAN (43.20)	Acoustic MoEKAN (5.02)	Acoustic MoEKAN (5.34)

TABLE VIII

BEST NON-KAN BASELINE PER FREQUENCY BAND

Frequency band (Hz)	Best non-KAN (MSE)	Best non-KAN (MAE)	Best non-KAN (MAPE)
0	TimesNet (122.22)	TimesNet (7.85)	TimesNet (13.57)
43.07~5512.50	RLinear (33.41)	STID (4.25)	STID (9.01)
5555.57~11025.00	STID (70.57)	STID (6.28)	STID (7.87)
11068.07~16537.50	RLinear (80.41)	RLinear (6.71)	RLinear (8.76)
16580.57~22050.00	iTransformer (46.09)	iTransformer (5.06)	iTransformer (5.45)

Table IX reports a controlled configuration comparison that varies (i) the number of experts ($E=4$ vs. $E=10$), (ii) the gating mechanism (dense Transformer gating vs. MoBA sparse routing), and (iii) the use of RevIN. Several observations follow. First, RevIN consistently improves robustness across all configurations, reducing AVG MSE by 3.6–4.2 ($\approx 5\%$) and yielding parallel reductions in AVG MAE/MAPE, which supports the role of window-wise distribution correction in STFT-domain forecasting. Second, expanding the expert pool from four to ten experts improves accuracy even under the same dense gate ($C1 \rightarrow C5$: AVG MSE 75.66 $\rightarrow$ 67.66; $C2 \rightarrow C6$: 79.86 $\rightarrow$ 71.26), indicating that the additional acoustic-motivated experts provide complementary function-space coverage beyond the base analytic set. Third, replacing dense gating with MoBA sparse routing yields modest gains when $E=4$ ($C1 \rightarrow C3$: 75.66 $\rightarrow$ 72.16), but becomes substantially more effective when $E=10$ ($C5 \rightarrow C7$: 67.66 $\rightarrow$ 60.06), suggesting that sparse top- k routing better exploits a larger expert pool by encouraging specialization while keeping the parameter count nearly unchanged within the same E ($\Delta\text{Params} \leq 0.33\text{M}$ between dense vs. MoBA). Overall, the full configuration ($E=10$ + MoBA + RevIN; $C7$) achieves the best average accuracy (60.06/5.62/8.23), confirming that the gains of AcousticMoEKAN arise from the combined effect of expert augmentation, sparse routing, and normalization rather than from a single engineering choice.

Finally, the KAN-based family (DenseRMoK, MoEKAN, and AcousticMoEKAN) tends to jointly approximate the nonlinear peak shape (rapid rise followed by decay) and the inter-peak floor. As illustrated in Fig. 14, the proposed AcousticMoEKAN most consistently reproduces both the timing and amplitude of peaks across representative bins. We attribute this to the combination of (i) an expanded expert pool and (ii) MoBA-based gating, which enables the model to select appropriate expert compositions for different input patterns (e.g., low-frequency trends vs. high-frequency impulses/reverberation), thereby mitigating both oversmoothing in spike regions and bias accumulation in floor regions.

1) Linear models (NLinear and RLinear). Linear models approximate future values of each frequency-bin as a linear combination of past observations, yielding a simple architecture with stable training dynamics. In the low-frequency band (43.07–5512.50 Hz), RLinear achieves a notably low MSE of 33.41, indicating that low-frequency components are relatively smooth (less affected by noise than high frequencies) and can be forecast to a certain extent via linear extrapolation. However, errors increase in the mid-frequency region ($\approx 5.5\sim 16.5$ kHz), where the complex nonlinear and non-stationary patterns induced by tap-changing cannot be sufficiently represented by a purely linear mapping. The consistent improvement of RLinear over NLinear (AVG MSE 99.31 $\rightarrow$ 80.15) suggests that normalization (e.g., RevIN) and/or residual design better compensates for distribution shifts and provides a learnable scale for the mapping.

2) GNN models (STID and TimeGNN). GNN-based forecasting models capture inter-variable relationships through a graph structure. However, interactions among frequency-bins are inherently continuous across the spectrum and are difficult to capture with a sparse and stable set of graph edges. Despite its relatively simple design that relies on node/time ID embeddings rather than explicit message passing, STID achieves strong overall performance (AVG MSE 80.43, 4th overall; AVG MAE 6.31, 3rd overall). This suggests that when each bin exhibits a relatively stable “spectral identity” (e.g., mean level, variation range, and event response), ID embeddings can serve as effective explanatory variables. In contrast, TimeGNN employs a more complex framework involving dynamic graph learning/sampling; if the relational structure is unstable or noisy, graph estimation may amplify errors, which may explain the consistently large errors observed for TimeGNN in this setting (AVG MSE 386.93).

3) CNN models (SCINet and TimesNet). CNN-based models are effective at capturing local patterns via convolutional filters, but long-horizon forecasting can be sensitive to receptive-field design, especially when event-driven rise-and-decay dynamics extend over time. TimesNet, which estimates dominant periods via FFT and learns 2D variations, achieves the best performance among non-KAN models at DC (0 Hz) (MSE 122.22, MAE 7.85).

TABLE IX
EFFECTS OF EXPERT POOL SIZE, GATING STRATEGY, AND REVIN ON OVERALL AVERAGE FORECASTING ERRORS

Case ID	Experts	Gate	RevIN	AVG MSE	AVG MAE	AVG MAPE	Params(M)
C1	4	Dense (Transformer)	On	75.66	6.21	9.31	15.67
C2	4	Dense (Transformer)	Off	79.86	6.49	9.81	15.67
C3	4	MoBA Sparse	On	72.16	6.09	9.09	15.90
C4	4	MoBA Sparse	Off	76.36	6.37	9.59	15.90
C5	10	Dense (Transformer)	On	67.66	5.94	8.81	52.90
C6	10	Dense (Transformer)	Off	71.26	6.16	9.21	52.90
C7	10	MoBA Sparse (AcousticMoEKAN)	On	60.06	5.62	8.23	53.23
C8	10	MoBA Sparse	Off	63.66	5.84	8.63	53.23



FIGURE 14. Qualitative Forecast Comparisons on the Test Set for a Representative Sample Segment at Several Frequency Bins (0.00 Hz (DC), 43.07 Hz, 5555.57 Hz, 11068.07 Hz, and 16580.57 Hz). Blue Denotes the Ground Truth and Orange Denotes the Prediction.

This implies that when the DC component behaves as a combination of smooth trends and low-frequency fluctuations, a 2D periodic representation can be advantageous. In contrast, SCINet's split-and-recombine multi-scale mechanism can accumulate errors if the non-stationary event structure is not decomposed stably; correspondingly, SCINet exhibits relatively poor overall performance (AVG MSE 260.05).

4) Transformer models (PatchTST and iTransformer). Transformers can directly attend to global dependencies within long input sequences, which is beneficial for learning long-context correlations across time and across frequency-bins. PatchTST summarizes inputs into patch-level tokens, reducing attention cost while retaining local patterns, and achieves an AVG MSE of 98.22. iTransformer "inverts" the tokenization by treating

variables (channels) as tokens, enabling efficient modeling of cross-variable correlations; it achieves the best performance among non-KAN models in the high-frequency band (16.58~22.05 kHz) (MSE 46.09, MAE 5.06, MAPE 5.45). This suggests that, even when per-bin SNR degrades at high frequencies, exploiting shared patterns across bins (e.g., common noise-floor variations and broadband energy distributions) via variable-wise correlations can be advantageous.

5) KAN models (DenseRMoK, MoEKAN, and AcousticMoEKAN). KAN-based models endow each connection (edge) with a learnable 1D function (e.g., B-splines), providing an adaptive and comparatively interpretable mechanism for approximating complex nonlinear functions. DenseRMoK employs a mixture of KAN experts but yields a lower overall performance (AVG

MSE 118.54) than the MoEKAN variants in this dataset. Potential reasons include: (i) insufficiently specialized expert roles, (ii) interference among experts due to dense routing, and/or (iii) limited ability to reflect the multi-scale characteristics of spectral events.

MoEKAN leverages four KAN experts (Taylor, Wavelet, Jacobi, and Fourier) to decompose patterns into trends (polynomial), localized spikes (wavelet), numerically stable curve approximation (orthogonal polynomials), and periodic/harmonic components (Fourier), and a Transformer-based gate blends expert outputs according to input characteristics. This design yields a clear improvement over non-KAN baselines (AVG MSE 75.66). Building on this foundation, AcousticMoEKAN (i) expands the expert pool from 4 to 10 to increase representational capacity and (ii) replaces dense Transformer gating with MoBA-based sparse routing, enhancing both sparsity and diversity in expert selection. This combination encourages finer-grained specialization across frequency-dependent patterns and, in particular, reduces large errors in the mid-to-high frequency bands.

In summary, the experimental results indicate that (1) in the low-frequency region, linearity and normalization can constitute strong baselines; (2) as frequency increases, capturing nonlinear and non-stationary event dynamics requires greater representational power and stable routing; and (3) AcousticMoEKAN's "expert expansion + MoBA gating" best matches these requirements, delivering consistent gains across all frequency bands.

VI. CONCLUSION

This paper presents AcousticMoEKAN, an STFT-based OLTC acoustic spectrogram forecasting framework that serves as a practical alternative to direct raw-waveform prediction at audio sampling rates. By modeling the temporal evolution of 513 frequency-bin trajectories, the proposed approach enables long-horizon forecasting and frequency-resolved analysis of state changes. AcousticMoEKAN extends MoEKAN's mixture-of-KAN architecture by increasing the number of experts from 4 to 10 and replacing dense Transformer gating with MoBA-based sparse routing. On a large-scale accelerated-aging dataset comprising 195,994 tap-changing events acquired at 44.1 kHz, AcousticMoEKAN achieved the lowest MSE, MAE, and MAPE among all compared baselines and delivered consistent improvements across most frequency bands. Future work will extend evaluation to oil-immersed and in-service OLTC conditions, investigate domain adaptation across acoustic propagation media, and develop downstream anomaly-scoring methods that link prediction deviations to specific fault modes. In addition, joint magnitude-phase prediction for time-domain reconstruction constitutes a promising direction when waveform recovery is required.

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